MEHUL PANDEY

AI Developer / AI Engineer

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SUMMARY

AI Developer with over a year of experience building production ML and LLM systems - document and vision pipelines, search ranking, fine-tuned language models, and agentic automation. I like taking a problem from raw data through to a working pipeline myself, and I'd rather ship something simple that hits the accuracy bar than spend weeks chasing a fancier model that barely moves the needle.

SKILLS

Languages & Core: Python, SQL, Git

ML / DL: Random Forest, Logistic Regression, scikit-learn, Transformers (BERT), model fine-tuning, model evaluation, statistical testing (t-tests), feature engineering

LLMs & GenAI: LLM fine-tuning (private 3B-parameter model), Hugging Face Transformers, LLM APIs, prompt engineering, VLM-based document extraction, agentic workflow design

NLP / CV: OCR, template-based document parsing, text classification, semantic search, embeddings

Backend & Infra: FastAPI, REST APIs, Docker, PostgreSQL, CI/CD (Azure DevOps), model deployment, ML pipelines

Tools: n8n-style workflow builders, Excel, Google ADK-style agent orchestration

EXPERIENCE

AI Developer — Toss Solutions *Mar 2026 – Present*

- Developed AI pipelines for e-commerce product mapping across multiple platforms and product categories. Exposed via FastAPI, backed by PostgreSQL, and containerized with Docker.
- Led development of a digital answer-booklet evaluation system: it auto-generates a grading rubric from the question paper, scores booklets against it, and allows manual override where needed. Added a feedback loop so accuracy keeps improving over time. Deployed via FastAPI, PostgreSQL, and Docker, with CI/CD through Azure DevOps. Evaluation accuracy hit ~85%+ across English, Hindi, and Mathematics.
- Built a semantic search and ranking pipeline using NDCG and embeddings for relevance mapping. Reached ~70-75% accuracy classifying results as excellent vs. non-excellent.
- Designed an in-house n8n-style workflow platform to connect external LLM APIs into the pipelines above, so the team could orchestrate them without writing custom glue code for each one.
- Built a VLM-based OCR tool that pulls structured fields from template-based official certificates into Excel. Also fine-tuned a private 3B-parameter transformer-based LLM for an internal use case.

Associate Software Consultant — 4th Orbit Pvt Ltd *Jul 2025 – Feb 2026*

- Built core modules of Gaplytiq, an EdTech platform. This included an OCR-based resume builder (10 templates), a resume-to-JD gap analysis engine that returns ATS-style match scores with a skill-gap report, a learning roadmap generator, a multi-format skill test engine, and an adaptive mock interview bot that generates and adjusts questions from a JD and resume using external LLM APIs.

Research Intern — IIST, Trivandrum *Mar 2024 – Jul 2024*

- Studied whether VR environments reduce stress compared to real-world conditions, using ECG data from 40 participants (RMSSD, HF power, NN50, LF/HF ratio). Applied Random Forest and Logistic Regression in scikit-learn to classify stress states, and validated the results with t-tests.

PROJECTS

Route Optimization System — Hack-a-Sol Hackathon, IIIT Naya Raipur

- Built a graph-based pathfinding solution over a road network to catch nodal points with no viable route and generate alternative paths on the fly. Later refined the routing logic to deprioritize poor-condition roads, which improved both route quality and how the map looked to end users.

BERT-Based Anxiety Detection System

- Took the ECG dataset from my IIST internship (baseline and stress conditions) and extended it with daily activity-tracking and weather data to build a broader feature set. Fine-tuned a Hugging Face BERT transformer on top of it to classify anxiety states — a side project to test whether physiological signals alone miss real-world stress factors.

EDUCATION & CERTIFICATIONS

B.Tech (Honors), Computer Science Engineering — Data Science CSVTU, Bhilai — CGPA 8.26

NPTEL: High Performance Scientific Computing, AI for Investments, Management Information System, Algorithmic Game Theory | Deloitte Australia Data Analytics Job Simulation (Forage)

ACHIEVEMENTS & LEADERSHIP

- 1st Place — Extended AI Competition, KK Modi University.
- PR & Communication Head, TPO Cell — CSVTU | Core Member, Aarohan Cultural Committee.